

Sudhanshu Srivastava

DATA SCIENTIST | FORECASTING | EXPERIMENTATION | PREDICTIVE MODELING

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AVAILABLE IMMEDIATELY | NO EMPLOYMENT SPONSORSHIP REQUIRED

SUMMARY

Data Scientist with 7+ years building and deploying forecasting, classification, experimentation, and optimization solutions from operational and customer data. Improved demand-forecast MAPE from about 22% to 13% across 500+ categories, raised risk-model recall from 68% to 87%, and reduced AWS model refresh from 2 days to 3 hours. PhD candidate (2026), available for full-time work immediately.

CORE SKILLS

Analytics: Forecasting, regression, classification, clustering, A/B testing, causal inference, experimental design, feature engineering, SHAP

Tools: Python, SQL, R, pandas, NumPy, SciPy, scikit-learn, XGBoost, PyTorch, TensorFlow, MLflow, Databricks

Data & Cloud: AWS SageMaker, AWS Glue, ETL pipelines, model monitoring, KPI dashboards, Git, Linux

PROFESSIONAL EXPERIENCE

Research Scientist | IBM - UofL | Louisville, KY

Aug 2024 - Present

- Built and deployed a causal ML system for semiconductor defect root-cause analysis using 5,173 industrial records; reduced causal-graph error (structural Hamming distance) 62.5% vs. 8 baselines and achieved 96.1% prediction accuracy.
- Designed and evaluated a stochastic production-scheduling model that delivered 98.5% job completion and outperformed GA+CPLEX by up to 53 percentage points, supporting real-time operational decisions.

AI Consultant | Hope and Dreams Trust | India

Feb 2024 - Jun 2024

- Developed an XGBoost early-risk model on 50K+ engagement records, raising recall from 68% to 87%; engineered behavioral features in SQL and validated performance with cross-validation and SHAP.
- Automated SQL, AWS Glue, SageMaker, and MLflow workflows, cutting model refresh from 2 days to 3 hours and flagging a 7% accuracy drift within 24 hours; an NLP routing pipeline saved 18+ review hours weekly.

Applied AI Researcher | UPES Dehradun | India

Aug 2019 - Jan 2024

- Partnered with IIM Ahmedabad and Nykaa on live transaction data; reduced demand-forecast MAPE from about 22% to 13% across 500+ categories using gradient-boosted trees and statistical time-series features.
- Segmented 500K+ transactions into actionable customer cohorts for campaign retargeting and led 10 applied analytics projects from data preparation through experimentation and performance evaluation.

Data Scientist | Maven General Contracting L.L.C. | Abu Dhabi, UAE

Jun 2016 - Jul 2017

- Built predictive-maintenance models from sensor and maintenance data, achieving 92% accuracy and reducing downtime 18%; forecasting and optimization dashboards cut annual operating costs by about AED 3.5M.

SELECTED PROJECTS

Session Recommender & Experimentation Framework | Python, PyTorch, FastAPI, MLflow, A/B Testing

[GitHub](#)

- Built a production session-recommendation and experimentation workflow with BERT4Rec, frequentist and Bayesian tests, power analysis, MLflow, and FastAPI; the public benchmark reached HR@10 of 0.27 vs. 0.10 for a popularity baseline.

A/B Testing & Experimentation Toolkit | Python, SciPy, PyMC, Streamlit

[GitHub](#)

- Built a reusable experimentation toolkit with frequentist and Bayesian tests, confidence and credible intervals, power and sample-size analysis, and an interactive interface for uploaded datasets.

Customer Segmentation Dashboard | pandas, scikit-learn, K-Means, Plotly Dash

[GitHub](#)

- Built an end-to-end segmentation workflow with preprocessing, cluster validation, customer profiling, and interactive visualizations for campaign targeting and cohort analysis.

PUBLICATIONS

10+ Research publications spanning forecasting, causal ML, reinforcement learning, and manufacturing analytics | [Publication list](#)

AWARDS & LEADERSHIP

Trailblazers in Engineering Fellow, Purdue (2026) | Outstanding Service Award, ISE Dept., University of Louisville - top 2 among all graduate students (2026) | President, Indian Graduate Student Association, UofL (2025-present) - led 9-member board; organization won Outstanding New Organization (2026) | American Society for Quality (ASQ) Scholarship (2025) | NSF Travel Award (2025, 2026) | 3-Minute Thesis Runner-up (2025) | NSF I-Corps Entrepreneurial Lead, StressNav (patent pending)

EDUCATION

PhD Candidate, Industrial & Systems Engineering | University of Louisville

2026

MBA, Operations Management | UPES Dehradun

2025

MTech, Computer Science & Engineering | IIT Dhanbad

2019

BE, Computer Science & Engineering | BIT Mesra

2016